M3 Topic 1 End of Topic Assessment · ANSWER KEY · DETAILED (step-by-step, easy words)

Algebra I · Module 3, Topic 1: Linear Equations and Inequalities · SE pp. 27–30 · 11 questions

Quick key

#	SE p.	Answer
1	27	**B. y = 20x + 50**
2	27	**J. No solution**
3	28	**D. s < −1**
4	28	**x = 1**
5	28	**Infinitely many solutions**
6	28	**y = 2.80x + 6.75**
7	29	**$y = -\frac{5}{3}x + 3$**
8	29	**$r = \frac{C}{2\pi}$**
9	29	**m < −4 (open circle at −4, shade left)**
10	30	**y = 2x + 11**
11	30	**n = 17**

Big ideas used on this test

- Do the same thing to **both sides** to keep it balanced.
- **Distribute**: multiply the outside number by everything inside the ( ).
- Inequalities: multiply/divide by a **negative** → **flip** the sign.
- Same x-part, different numbers → **no solution**. Both sides identical → **infinite**.
- Slope-intercept form: **y = mx + b** (m = rate/slope, b = starting value).

Question 1

SE p. 27

Sofia: $50 from her aunt, saves $20 per week. Equation for amount y after x weeks?

B. y = 20x + 50

1. Sofia **starts** with $50. That money is there on day one, so it is the number that is added by itself: **+ 50**.
2. Every week she adds $20. "Per week" means it gets multiplied by the number of weeks: **20 × x = 20x**.
3. Put them together: money = 20x + 50, so **y = 20x + 50**.
4. Check with a real week: after 2 weeks she has 50 + 20 + 20 = $90. The formula gives 20(2) + 50 = 90 ✓.
5. Choice **A** swaps the numbers (she does not save $50 a week). **C** and **D** have a minus sign, but her money is going **up**, not down.

Question 2

SE p. 27

What value of p makes 8 − 6(p + 4) = 12 − 2(3p + 10) true?

J. No solution

1. **Distribute** means multiply the number outside the parentheses by each thing inside. Left side: −6 × p = −6p and −6 × 4 = −24. So 8 − 6p − 24.
2. Right side: −2 × 3p = −6p and −2 × 10 = −20. So 12 − 6p − 20.
3. Put the plain numbers together on each side: 8 − 24 = −16 and 12 − 20 = −8. Now it says **−6p − 16 = −6p − 8**.
4. Both sides have −6p. Add 6p to both sides and the p's disappear: **−16 = −8**.
5. −16 is NOT equal to −8. That is a false statement, so no number for p can ever work. Answer: **No solution**.
6. Tip: same p-part on both sides + different numbers = no solution.

Question 3

SE p. 28

Solution set for 4(2 − s) > −3s + 9?

D. s < −1

1. Distribute the 4: 4 × 2 = 8 and 4 × (−s) = −4s. Now: **8 − 4s > −3s + 9**.
2. Get the s's on one side. Add 3s to both sides: −4s + 3s = −s. Now: **8 − s > 9**.
3. Get the numbers on the other side. Subtract 8 from both sides: **−s > 1**.
4. We want s, not −s, so divide both sides by −1. **Rule: when you multiply or divide by a negative number, flip the sign.** > becomes <.
5. Answer: **s < −1**.
6. Check with a number that should work, s = −2: 4(2 − (−2)) = 4(4) = 16, and −3(−2) + 9 = 6 + 9 = 15. Is 16 > 15? Yes ✓.

Question 4

SE p. 28

Solve 0.4(5x − 2) = −2.4x + 3.6

x = 1

1. Distribute 0.4: 0.4 × 5x = 2x (because 0.4 × 5 = 2) and 0.4 × 2 = 0.8. Now: **2x − 0.8 = −2.4x + 3.6**.
2. Get all the x's on the left. Add 2.4x to both sides: 2x + 2.4x = 4.4x. Now: **4.4x − 0.8 = 3.6**.
3. Get the numbers on the right. Add 0.8 to both sides: 3.6 + 0.8 = 4.4. Now: **4.4x = 4.4**.
4. 4.4 times what is 4.4? Divide both sides by 4.4: **x = 1**.
5. Check: left 0.4(5 − 2) = 0.4(3) = 1.2; right −2.4 + 3.6 = 1.2. Same ✓.

Question 5

SE p. 28

One solution, no solution, or infinite? $\frac{1}{3}(9x + 45) = 3(x + 3) + 6$

Infinitely many solutions

1. Left side: one-third of each part. 9x ÷ 3 = 3x and 45 ÷ 3 = 15. So the left side is **3x + 15**.
2. Right side: 3 × x = 3x and 3 × 3 = 9, then + 6. So 3x + 9 + 6 = **3x + 15**.
3. Now it says **3x + 15 = 3x + 15**. Both sides are exactly the same!
4. If you subtract 3x and 15 from both sides you get 0 = 0, which is always true. So **any** number for x works.
5. Answer: **infinite number of solutions**, because both sides simplify to the same expression, 3x + 15.
6. Try it: x = 0 gives 15 = 15 ✓, x = 10 gives 45 = 45 ✓.

Question 6

SE p. 28

Chloe buys 3 eclairs at $2.25 each and x doughnuts at $2.80 each. Equation for total cost y?

y = 2.80x + 6.75

1. The eclairs are always 3, so their cost never changes: 3 × $2.25 = **$6.75**. (2.25 + 2.25 + 2.25 = 6.75.)
2. We don't know how many doughnuts, so call it x. Each one is $2.80, so doughnuts cost **2.80 × x = 2.80x**.
3. Total = doughnuts + eclairs: **y = 2.80x + 6.75**.
4. Check: 2 doughnuts → 2.80(2) + 6.75 = 5.60 + 6.75 = $12.35 ✓.

Question 7

SE p. 29

Rewrite 5x + 3y = 9 in slope-intercept form.

$y = -\frac{5}{3}x + 3$

1. **Slope-intercept form** means y is all by itself: y = mx + b.
2. Move the 5x away from the y. Subtract 5x from both sides: **3y = −5x + 9**. (Write the x-part first.)
3. y is being multiplied by 3. Divide **everything** on both sides by 3: −5x ÷ 3 = $-\frac{5}{3}x$ and 9 ÷ 3 = 3.
4. Answer: **$y = -\frac{5}{3}x + 3$**. Slope is −5/3, y-intercept is 3.
5. Common mistake: dividing only the 9 by 3 and forgetting the −5x.

Question 8

SE p. 29

C = 2πr. Solve for r.

$r = \frac{C}{2\pi}$

1. We want r by itself. Right now r is being **multiplied** by 2 and by π.
2. To undo multiplying, we **divide**. Divide both sides by 2π.
3. On the right, 2πr ÷ 2π leaves just r. On the left we get C ÷ 2π.
4. Answer: **$r = \frac{C}{2\pi}$** (same as r = C ÷ (2π)).
5. Like a number problem: if 10 = 2 × 5, then 5 = 10 ÷ 2.

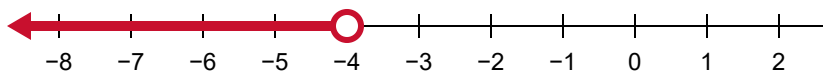
Question 9

SE p. 29

Solve 1.5m − 2 < −2(m + 8) and graph on the number line.

m < −4 (open circle at −4, shade left)

1. Distribute −2: −2 × m = −2m and −2 × 8 = −16. Now: **1.5m − 2 < −2m − 16**.
2. Get the m's on the left. Add 2m to both sides: 1.5m + 2m = 3.5m. Now: **3.5m − 2 < −16**.
3. Get numbers on the right. Add 2 to both sides: −16 + 2 = −14. Now: **3.5m < −14**.
4. Divide both sides by 3.5: −14 ÷ 3.5 = −4. We divided by a **positive** number, so the sign stays the same: **m < −4**.
5. Graph: **open** (empty) circle on −4, because −4 itself does not work (it's < , not ≤). Shade to the **left**, where the smaller numbers are.



6. Check: m = −5 → 1.5(−5) − 2 = −9.5 and −2(−5 + 8) = −6. Is −9.5 < −6? Yes ✓.

Question 10

SE p. 30

Rewrite y − 5 = 2(x + 3) in slope-intercept form.

y = 2x + 11

1. Distribute the 2: 2 × x = 2x and 2 × 3 = 6. Now: **y − 5 = 2x + 6**.
2. y has a −5 with it. Add 5 to both sides to get y alone: 6 + 5 = 11.
3. Answer: **y = 2x + 11**. Slope 2, y-intercept 11.
4. Check: x = 0 → original: y − 5 = 2(3) = 6, so y = 11 ✓.

Question 11

SE p. 30

Triangle: height (2n − 4) in, base 8 in. Rectangle: length 6 in, width (3 + n) in. Areas equal. Find n.

n = 17

1. **Area of a triangle** = ½ × base × height. Half of 8 is 4, so the area is 4 × (2n − 4) = **8n − 16**.
2. **Area of a rectangle** = length × width = 6 × (3 + n) = **18 + 6n**.
3. The problem says the areas are equal, so set them equal: **8n − 16 = 6n + 18**.
4. Subtract 6n from both sides: **2n − 16 = 18**. Add 16 to both sides: **2n = 34**.
5. Divide by 2: **n = 17**.
6. Check: triangle height 2(17) − 4 = 30, area ½ × 8 × 30 = 120. Rectangle width 3 + 17 = 20, area 6 × 20 = 120. Equal ✓.